Phishing-Email Analysis Report

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1. What is the Phishing-email?

- "Phishing" refers to an attempt to steal sensitive information, typically in the form of usernames, passwords, credit card numbers, bank account information, or other important data in order to utilize or sell the stolen information.
- Phishing attacks aim to steal or damage sensitive data by deceiving people into revealing personal information like passwords and credit card numbers.

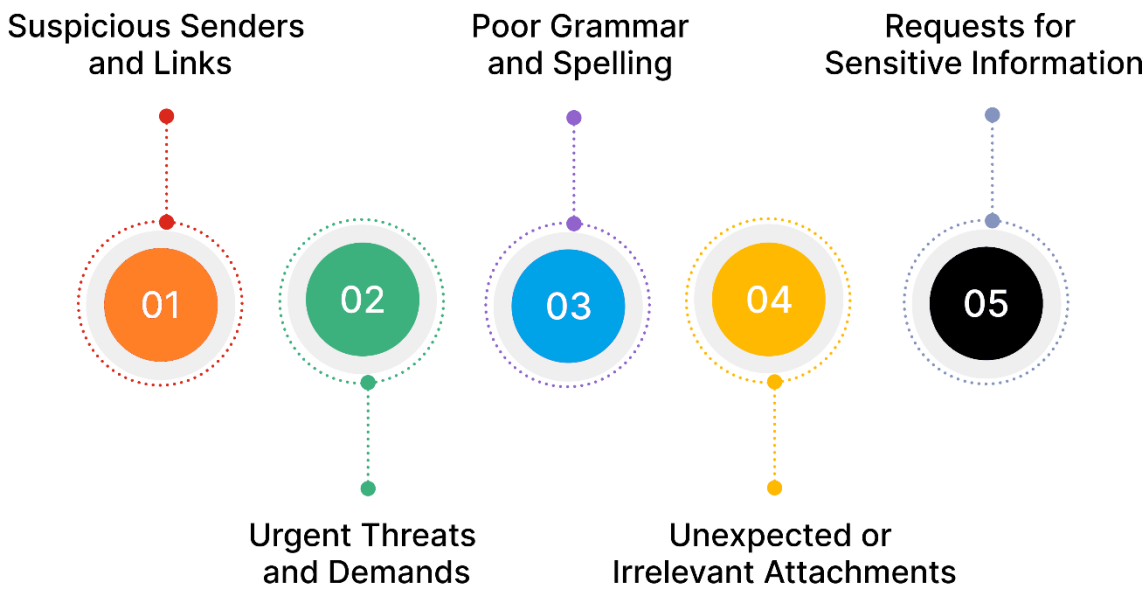
2. Phishing Email Types:

- **Email Phishing:** Attackers send fake emails to many people.
- **Spear Phishing:** A targeted phishing attack on a specific person or organization.
- **Whaling:** Phishing attack targeting high-level people.
- **Clone Phishing:** A real email is copied, but the link or attachment is changed to a malicious one.
- **Smishing:** Phishing done through SMS messages.
- **Vishing:** Phishing using voice calls.

3. Phishing Indicators:

- There are some points (indictors) which is help to identify the phishing-email.
    1. Fake email address (example: support@paypa1.com instead of paypal.com)
    2. Suspicious attachments (.exe, .zip, .html files)
    3. Links that redirect to fake websites
    4. Mismatched sender domain
    5. Unexpected login alerts
    6. Email spoofing

4. Five Main Red Flags to identify Phishing Emails:



5. Phishing Techniques Used:

- Explain techniques used by attackers.

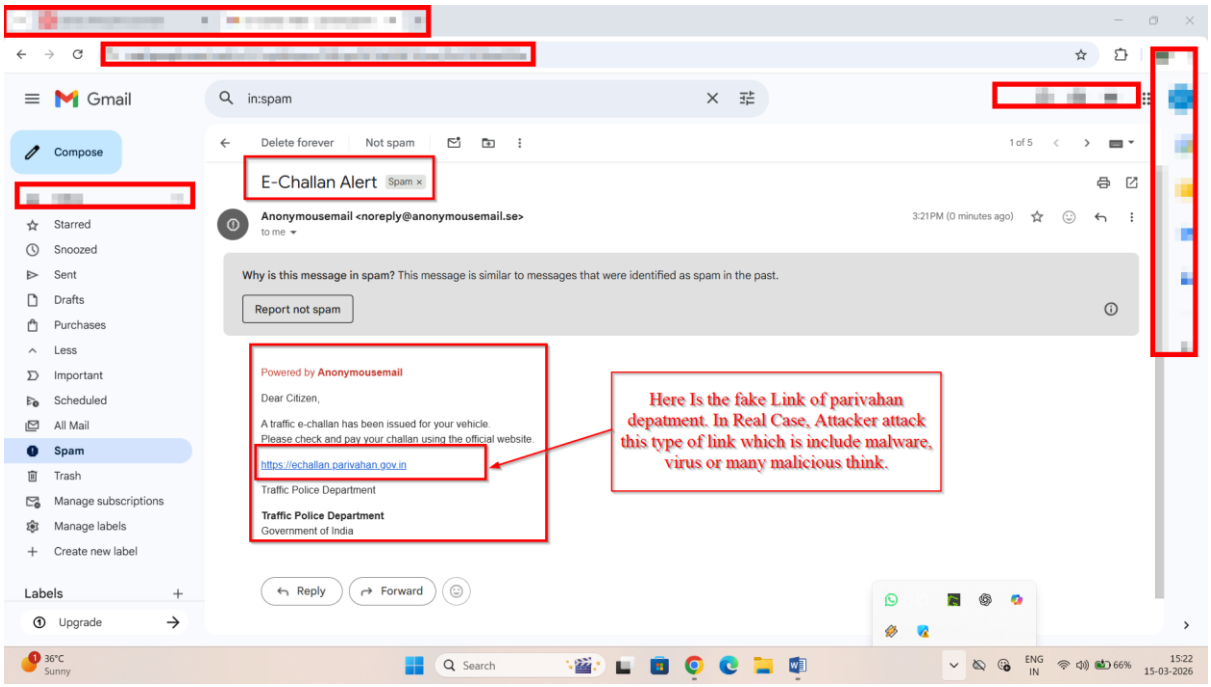
1. **Email Spoofing:** Attackers send emails that look like they come from a trusted person or organization.
2. **Brand Impersonation:** Attackers pretend to be popular companies like banks or online services to gain trust.
3. **Fake Login Pages:** Phishing emails contain links to fake websites that steal usernames and passwords.
4. **Urgency and Fear Tactics:** Attackers create panic by saying the account will be locked or suspended.
5. **Malicious Links:** Emails contain harmful links that lead to phishing or malware websites.
6. **Malicious Attachments:** Some phishing emails include infected files that can install malware on the device.
7. **Social Engineering:** Attackers trick people by using trust, fear, or curiosity to get sensitive information.

6. Prevention Methods:

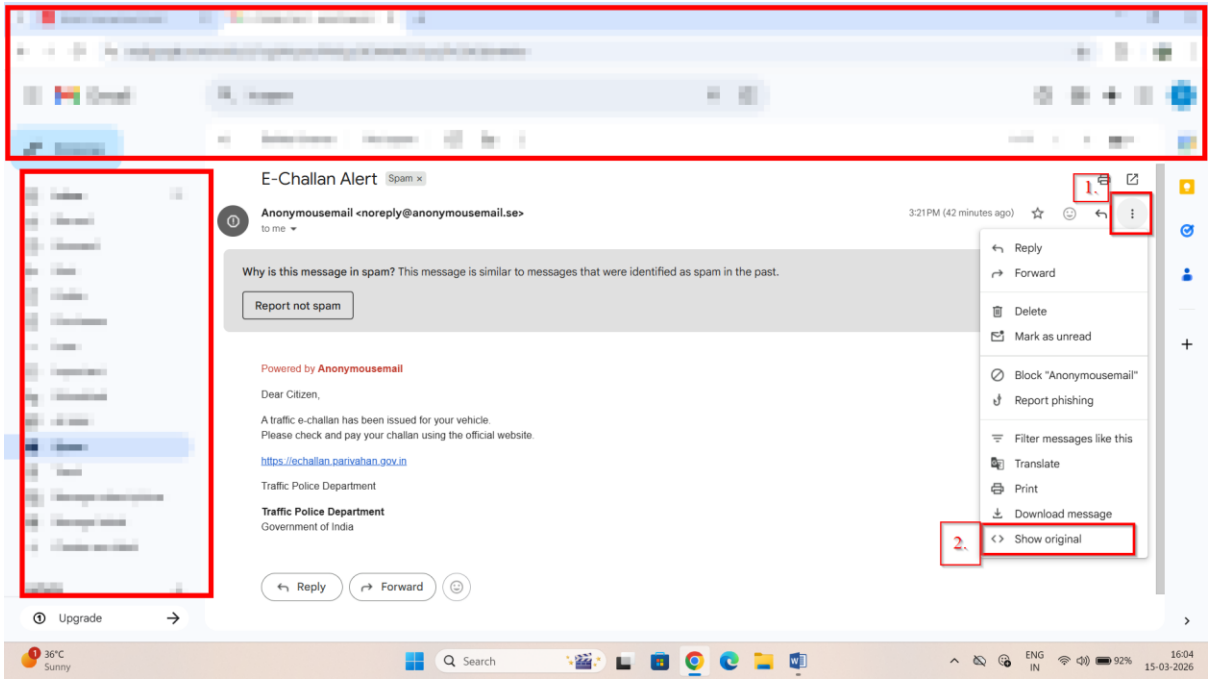
- **Verify Sender and Links:** Always check the sender's email address for slight misspellings. Hover your mouse over links to preview the actual destination URL before clicking.
- **Avoid Urgent Requests:** Be wary of emails demanding immediate action, such as account suspension threats, which are designed to create panic.
- **Use Multi-Factor Authentication (MFA):** Enable MFA on all accounts, as it acts as a crucial barrier even if a password is stolen.
- **Keep Software Updated:** Regularly update web browsers, antivirus software, and operating systems to patch security vulnerabilities.
- **Use Password Managers:** Password managers can prevent phishing by not auto-filling credentials on lookalike or malicious websites.
- **Deploy Email Security Tools:** Use anti-phishing software, SPF, DKIM, and DMARC protocols to filter spam and verify sender identity.
- **Educate and Train:** Implement security awareness training to help recognize phishing attempts.

7. Practical Demonstration:

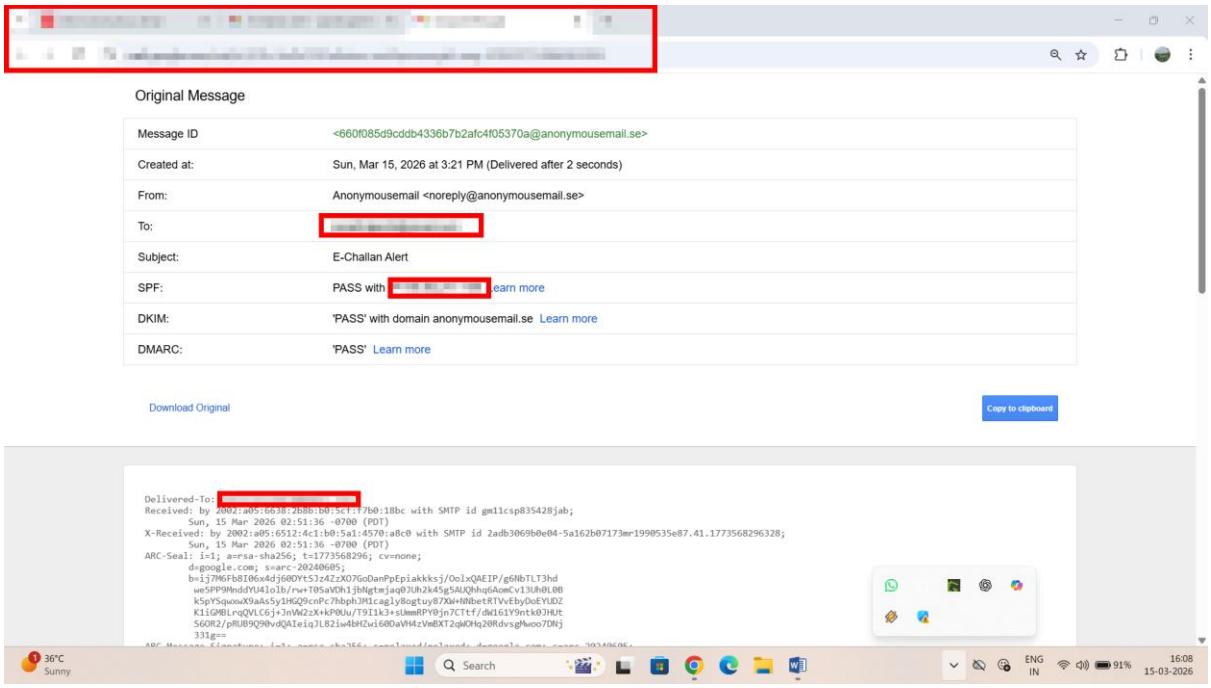
- Here, phishing email claiming to be an e-challan from the Government Parivahan Department.



- Now we analyse this email using the following steps.



- After follow that process then show all details of that e-mail.
- Here show in header, all information of email like To, From, Subject, Security Keys.



8. E-mail Analysis:

- Here Delivered phishing email received in the inbox.

```
Delivered-To: [redacted]@gmail.com
```

- Here Email header showing the delivery path and protocols used.

```
Received: by 2002:a05:6638:2b8b:b0:5cf:f7b0:18bc with SMTP id gm11csp835428jab;
```

- Here, show the timing and date of receive e-mail.

```
X-Received: by 2002:a05:6512:4c1:b0:5a1:4570:a8c0 with SMTP id 2adb3069b0e04-5a162b07173mr1990535e87.41.1773568296328;
        Sun, 15 Mar 2026 02:51:36 -0700 (PDT)
```

- DKIM is an email authentication method used to verify the sender. If DKIM does not pass, the email signature may be invalid or the email may be spoofed. SPF and DMARC are also email authentication mechanisms used to verify the legitimacy of the sender.
- Email authentication results showing SPF, DKIM, and DMARC verification.
- **SPF** verifies the sending server
- **DKIM**: verifies the email signature/content
- **DMARC:** enforces policies using SPF and DKIM results

```
dkim=pass header.i=@anonymousemail.se header.s=mail header.b=ajcSN52i;
spf=pass (google.com: domain of noreply@anonymousemail.se designates 185.82.217.109 as permitted sender) smtp.mailfrom=noreply@anonymousemail.se;
dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=anonymousemail.se
```

- Here **Return-Path** that is show the real sender address used for email delivery. **s3.inboxpress.net** is show the mail server that sent the email and also show the IP address which is sending server.

```
Return-Path: <noreply@anonymousemail.se>
Received: from s3.inboxpress.net (s3.inboxpress.net. [185.82.217.109])
        by mx.google.com with ESMTPS id 38308e7fff4ca-38a67d96190si41756051fa.126.2026.03.15.02.51.35
        for [redacted]
        (version=TLS1_3 cipher=TLS_AES_256_GCM_SHA384 bits=256/256);
        Sun, 15 Mar 2026 02:51:36 -0700 (PDT)
```

- The DKIM, SPF, and DMARC checks passed, confirming that the sending server (185.82.217.109) is authorized to send emails for anonymousemail.se.
- The SMTP envelope sender address used to send the email is noreply@anonymousemail.se.

```
Received-SPF: pass (google.com: domain of noreply@anonymousemail.se designates 185.82.217.109 as permitted sender) client-ip=185.82.217.109;
Authentication-Results: mx.google.com;
       dkim=pass header.i=@anonymousemail.se header.s=mail header.b=ajcSN52i;
       spf=pass (google.com: domain of noreply@anonymousemail.se designates 185.82.217.109 as permitted sender) smtp.mailfrom=noreply@anonymousemail.se;
       dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=anonymousemail.se
```

- **From Address:** The email appears to be sent from Anonymousemail with the address noreply@anonymousemail.se , which is the visible sender of the message.

- **Subject:** The subject of the email is "E-Challan Alert," which is used to inform the recipient about a traffic challan.

- **Message-ID**: The Message-ID is a unique identifier for the email, and it shows that the message was generated by the anonymousemail.se mail system.

- **Email Content Observation**: The email claims to be from the Traffic Police Department, Government of India, but the sender domain is anonymousemail.se. Official government emails should normally come from.gov.in domains, so this difference is suspicious.

```
09:51:35 +0000 (UTC)
Date: Sun, 15 Mar 2026 10:51:34 +0100
To: [redacted]
From: Anonymousemail <noreply@anonymousemail.se>
Subject: E-Challan Alert
Message-ID: <660f085d9cddb4336b7b2afc4f05370a@anonymousemail.se>
List-Unsubscribe-Post: List-Unsubscribe=One-Click
List-Unsubscribe: <https://anonymousemail.se/unsubscribe.php?uid=Nlh0VVIrdHE2K2VHMVZweWE4NXNFb3lyY0RVb1R3S1lRR053WjQybU8vWT0=>
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="b1=_IZKfEUEBJx52lz8vVUBv8AVRw6ABnqbnZaSzsTqqg"
Content-Transfer-Encoding: 8bit

--b1=_IZKfEUEBJx52lz8vVUBv8AVRw6ABnqbnZaSzsTqqg
Content-Type: text/plain; charset=us-ascii

Powered by Anonymousemail

Dear Citizen,

A traffic e-challan has been issued for your vehicle.
Please check and pay your challan using the official website.

https://echallan.parivahan.gov.in

Traffic Police Department

Traffic Police Department
Government of India
```

9. Conclusion: The analyzed email shows several phishing indicators. Although SPF, DKIM, and DMARC authentication checks passed, the email claims to be from the Government of India while the sender domain is anonymousemail.se. This mismatch indicates a potential phishing attempt designed to trick users into clicking the provided link.

Reference

- https://www.microsoft.com/en-us/security/business/security-101/what-is-phishing
- https://www.cloudflare.com/en-in/learning/access-management/phishing-attack/
- https://www.fortinet.com/resources/cyberglossary/phishing-email-analysis
- https://www.google.com/